**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

DAIRY FACILITY INFORMATION

**A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY:** John Zylstra Dairy

Physical address of dairy:

| Number and Street | City | County | Zip Code |
|---|---|---|---|
| 3307 S Blaker RD | Turlock | Stanislaus | 95380 |

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X058-X026-X001-XXXX X058-X026-X005-XXXX X058-X026-X006-XXXX X058-X027-X013-XXXX

**B. OPERATORS**

Zylstra, John

Operator name: Zylstra, John

Telephone no.: Landline: ; Cellular: (123) 000-0000

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 3307 S Blaker RD | Turlock | CA | 95380 |

**This operator is responsible for paying permit fees.**

**C. OWNERS**

Trust, John Zylstra, A&N 1999 Trust-Survivors

Legal owner name: Trust, John Zylstra, A&N 1999 Trust-Survivors

Telephone no.: Landline: ; Cellular: (123) 000-0000

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 3307 S Blaker RD | Turlock | CA | 95380 |

**Annual Report - General Order No. R5-2007-0035**
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AVAILABLE NUTRIENTS

## A. HERD INFORMATION

| | Milk Cows | Dry Cows | Bred Heifers (15-24 mo.) | Heifers (7-14 mo. to breeding) | Calves (4-6 mo.) | Calves (0-3 mo.) |
|---|---|---|---|---|---|---|
| Number open confinement | 0 | 20 | 30 | 50 | 15 | 35 |
| Number under roof | 250 | 0 | 0 | 0 | 0 | 0 |
| Maximum number | 250 | 20 | 30 | 50 | 15 | 35 |
| Average number | 250 | 20 | 30 | 50 | 15 | 35 |
| Avg live weight (lbs) | 1,400 | 1,450 | 900 | 600 | | |

Predominant milk cow breed: Holstein

Average milk production: 64 *pounds per cow per day*

## B. MANURE GENERATED

Total manure excreted by the herd: 7,462.07 *tons per reporting period*

Total nitrogen from manure: 95,340.33 *lbs per reporting period*

After ammonia losses (30% loss applied): 66,738.23 *lbs per reporting period*

Total phosphorus from manure: 15,718.52 *lbs per reporting period*

Total potassium from manure: 46,891.81 *lbs per reporting period*

Total salt from manure: 122,311.50 *lbs per reporting period*

## C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 2,044,800 *gallons*

Total nitrogen generated: 3,721.36 *lbs*

Total phosphorus generated: 1,089.97 *lbs*

Total potassium generated: 7,295.06 *lbs*

Total salt generated: 38,506.63 *lbs*

| | | |
|---|---|---|
| | 2,044,800 | *gallons applied* |
| + | 0 | *gallons exported* |
| - | 0 | *gallons imported* |
| = | 2,044,800 | *gallons generated* |

## D. FRESH WATER SOURCES

| Source Description | Type |
|---|---|
| Ag Well | Ground water |
| Canal Water | Surface water |
| Domestic Well | Ground water |

## E. SUBSURFACE (TILE) DRAINAGE SOURCES

*No subsurface (tile) drainage sources entered.*

## F. NUTRIENT IMPORTS

*No dry manure nutrient imports entered.*

*No process wastewater nutrient imports entered.*

| Date | Material type / Description | Quantity | Reporting basis | Moisture (%) | N (%) | P (%) | K (%) | Salt (%) |
|---|---|---|---|---|---|---|---|---|
| 05/15/2023 | Liquid commercial fertilizer<br>*4.5-10-10-3* | 1,250.00 *gal* | | | 4.000000 | 10.000000 | 10.000000 | 0.000000 |

| Material type | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Commercial fertilizer / Other | 417.25 | 1,043.13 | 1,043.13 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Total imports for all materials | 417.25 | 1,043.13 | 1,043.13 | 0.00 |

## G. NUTRIENT EXPORTS

*No solid nutrient exports entered.*

*No liquid nutrient exports entered.*

APPLICATION AREA

## A. LIST OF LAND APPLICATION AREAS

| Field name | Controlled acres | Cropable acres | Total harvests | Type of waste applied | Parcel number |
|---|---|---|---|---|---|
| 1 | 53 | 50 | 2 | process wastewater | X058-X026-X001-XXXX |
| 2 | 53 | 43 | 0 | none | X058-X026-X005-XXXX |
| 3 | 18 | 18 | 1 | manure | X058-X027-X013-XXXX |
| 5 | 9 | 9 | 0 | none | X058-X027-X018-XXXX |
| Morgan | 9 | 8 | 0 | none | X058-X007-X013-XXXX |
| Totals for areas that were used for application | 71 | 68 | 3 | | |
| Totals for areas that were not used for application | 71 | 60 | 0 | | |
| Land application area totals | 142 | 128 | 3 | | |

## B. CROPS AND HARVESTS

1

Field name: 1

11/25/2022: Winter Forage

Crop: Winter Forage Acres planted: 50 Plant date: 11/25/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/06/2023 | 750.00 *ton* | Dry-weight | | 78.4 | 25,400.00 | 2,590.00 | 15,400.00 | | 17.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 15.00 | 164.59 | 16.78 | 99.79 | 1,101.60 |

1

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 05/15/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/08/2023 | 1,500.00 *ton* | Dry-weight | | 0.1 | 0.00 | 0.00 | 0.00 | | 0.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 30.00 | 0.00 | 0.00 | 0.00 | 0.00 |

3

Field name: 3

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/15/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/08/2023 | 540.00 *ton* | Dry-weight | | 0.1 | 0.00 | 0.00 | 0.00 | | 0.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 30.00 | 0.00 | 0.00 | 0.00 | 0.00 |

NUTRIENT BUDGET

## A. LAND APPLICATIONS

### 1 - 11/25/2022: Winter Forage

Field name: 1

Crop: Winter Forage

Plant date: 11/25/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 26.03 | 6.73 | 55.88 | 266.77 | 432,000.00 *gal* |
| Ag Well | Ground water | 7.43 | 0.00 | 0.00 | 216.30 | 864,000.00 *gal* |
| Application event totals | | 33.45 | 6.73 | 55.88 | 483.08 | |

### 1 - 05/15/2023: Corn, silage

Field name: 1

Crop: Corn, silage

Plant date: 05/15/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 17.35 | 4.49 | 37.25 | 177.85 | 288,000.00 *gal* |
| Canal Water | Surface water | 20.82 | 0.00 | 0.00 | 485.42 | 6,463,168.80 *gal* |
| Application event totals | | 38.17 | 4.49 | 37.25 | 663.26 | |

| | | | | |
|---|---|---|---|---|
| 05/15/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| 4.5-10-10-3 | Liquid commercial fertilizer | 1.30 | 26.25 | 26.25 | 0.00 | |
| Application event totals | | 1.30 | 26.25 | 26.25 | 0.00 | |

1 - 05/15/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/15/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 11.07 | 3.52 | 17.71 | 103.83 | 345,600.00 *gal* |
| Canal Water | Surface water | 24.98 | 0.00 | 0.00 | 582.50 | 7,755,802.57 *gal* |
| Application event totals | | 36.06 | 3.52 | 17.71 | 686.32 | |

| | | | | |
|---|---|---|---|---|
| 06/27/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 6.46 | 2.05 | 10.33 | 60.56 | 201,600.00 *gal* |
| Canal Water | Surface water | 14.57 | 0.00 | 0.00 | 339.79 | 4,524,218.16 *gal* |
| Application event totals | | 21.03 | 2.05 | 10.33 | 400.36 | |

| | | | | |
|---|---|---|---|---|
| 07/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 6.46 | 2.05 | 10.33 | 60.56 | 201,600.00 *gal* |
| Canal Water | Surface water | 14.57 | 0.00 | 0.00 | 339.79 | 4,524,218.16 *gal* |
| Application event totals | | 21.03 | 2.05 | 10.33 | 400.36 | |

| | | | | |
|---|---|---|---|---|
| 07/21/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 6.46 | 2.05 | 10.33 | 60.56 | 201,600.00 *gal* |
| Canal Water | Surface water | 14.57 | 0.00 | 0.00 | 339.79 | 4,524,218.16 *gal* |
| Application event totals | | 21.03 | 2.05 | 10.33 | 400.36 | |

| | | | | |
|---|---|---|---|---|
| 08/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 0.32 | 0.48 | 2.19 | 21.53 | 201,600.00 *gal* |
| Canal Water | Surface water | 14.57 | 0.00 | 0.00 | 339.79 | 4,524,218.16 *gal* |
| Application event totals | | 14.89 | 0.48 | 2.19 | 361.33 | |

## 1 - 05/15/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/14/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 0.27 | 0.42 | 1.88 | 18.46 | 172,800.00 *gal* |
| Canal Water | Surface water | 12.49 | 0.00 | 0.00 | 291.25 | 3,877,901.28 *gal* |
| Application event totals | | 12.76 | 0.42 | 1.88 | 309.71 | |

## 3 - 05/15/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 05/15/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/11/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Manure | Corral solids | 164.44 | 44.76 | 247.75 | 0.00 | 180.00 *ton* |
| Application event totals | | 164.44 | 44.76 | 247.75 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Canal Water | Surface water | 21.69 | 0.00 | 0.00 | 505.64 | 2,423,688.30 *gal* |
| Application event totals | | 21.69 | 0.00 | 0.00 | 505.64 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Canal Water | Surface water | 21.69 | 0.00 | 0.00 | 505.64 | 2,423,688.30 *gal* |
| Application event totals | | 21.69 | 0.00 | 0.00 | 505.64 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Canal Water | Surface water | 21.69 | 0.00 | 0.00 | 505.64 | 2,423,688.30 *gal* |
| Application event totals | | 21.69 | 0.00 | 0.00 | 505.64 | |

3 - 05/15/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Canal Water | Surface water | 21.69 | 0.00 | 0.00 | 505.64 | 2,423,688.30 *gal* |
| Application event totals | | 21.69 | 0.00 | 0.00 | 505.64 | |

| | | | | |
|---|---|---|---|---|
| 07/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Canal Water | Surface water | 21.69 | 0.00 | 0.00 | 505.64 | 2,423,688.30 *gal* |
| Application event totals | | 21.69 | 0.00 | 0.00 | 505.64 | |

| | | | | |
|---|---|---|---|---|
| 08/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Canal Water | Surface water | 21.69 | 0.00 | 0.00 | 505.64 | 2,423,688.30 *gal* |
| Application event totals | | 21.69 | 0.00 | 0.00 | 505.64 | |

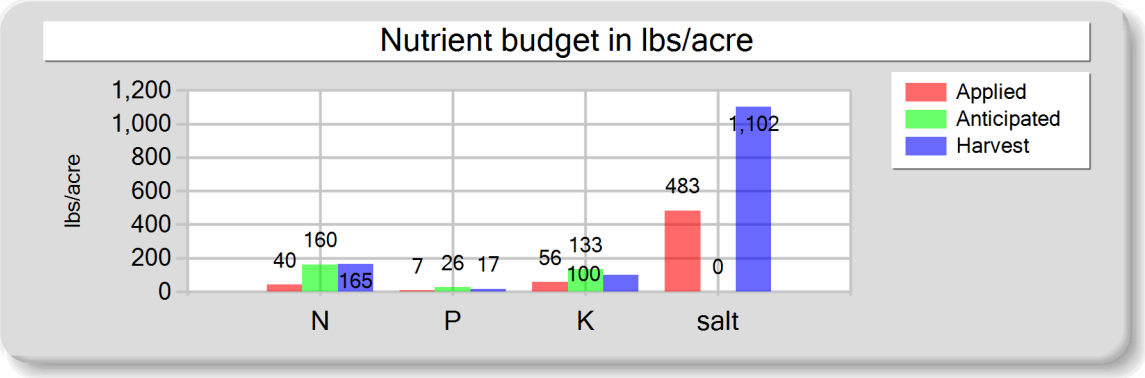
| | | | | |
|---|---|---|---|---|
| 08/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Canal Water | Surface water | 21.69 | 0.00 | 0.00 | 505.64 | 2,423,688.30 *gal* |
| Application event totals | | 21.69 | 0.00 | 0.00 | 505.64 | |

## B. NUTRIENT BUDGET

1 - 11/25/2022: Winter Forage

Field name: 1 Crop: Winter Forage Plant date: 11/25/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 26.03 | 6.73 | 55.88 | 266.77 |
| Fresh water | 7.43 | 0.00 | 0.00 | 216.30 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 40.45 | 6.73 | 55.88 | 483.08 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 164.59 | 16.78 | 99.79 | 1,101.60 |
| Nutrient balance | -124.14 | -10.05 | -43.91 | -618.52 |
| Applied to removed ratio | 0.25 | 0.40 | 0.56 | 0.44 |

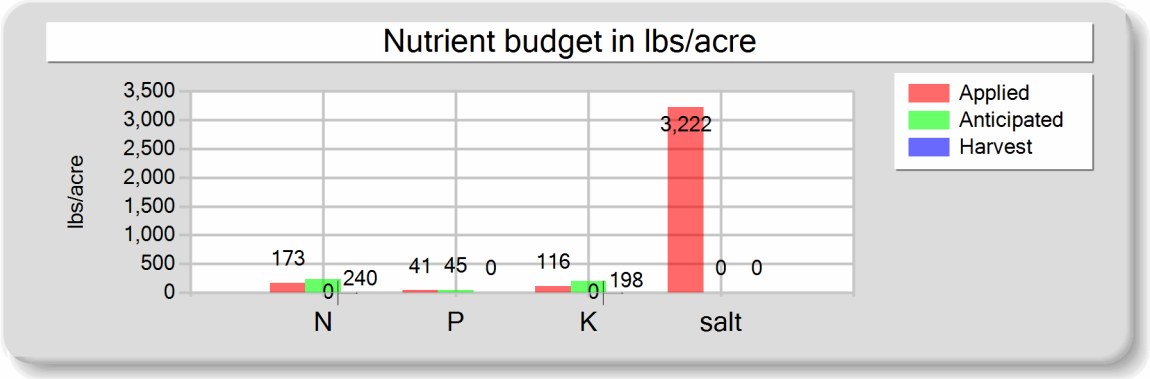
| Fresh water applied |
|---|
| 864,000.00 *gallons* |
| 31.82 *acre-inches* |
| 0.64 *inches/acre* |

| Process wastewater applied |
|---|
| 432,000.00 *gallons* |
| 15.91 *acre-inches* |
| 0.32 *inches/acre* |

| Total harvests for the crop |
|---|
| 1 *harvests* |

## 1 - 05/15/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/15/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 1.30 | 26.25 | 26.25 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 48.40 | 15.07 | 90.02 | 503.36 |
| Fresh water | 116.59 | 0.00 | 0.00 | 2,718.33 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 173.28 | 41.32 | 116.27 | 3,221.69 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 0.00 | 0.00 | 0.00 | 0.00 |
| Nutrient balance | 173.28 | 41.32 | 116.27 | 3,221.69 |
| Applied to removed ratio | 0.00 | 0.00 | 0.00 | 0.00 |

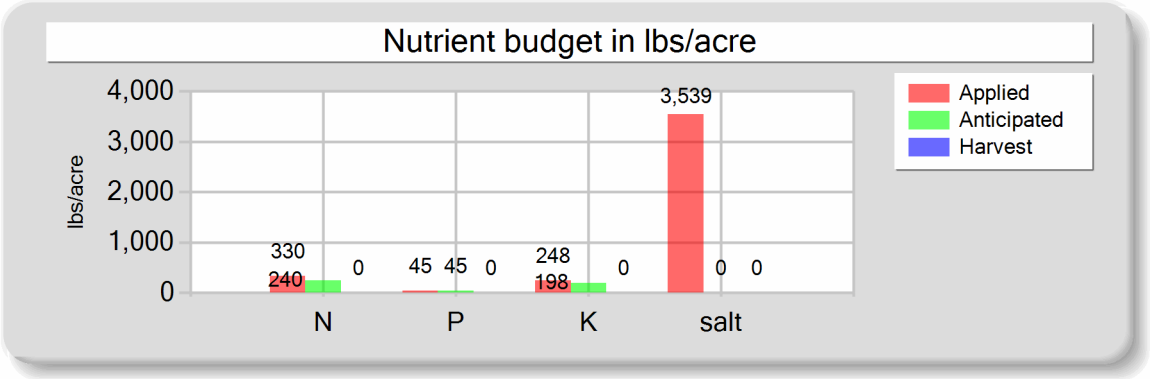
| Fresh water applied | |
|---|---|
| 36,193,745.29 | *gallons* |
| 1,332.89 | *acre-inches* |
| 26.66 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,612,800.00 | *gallons* |
| 59.39 | *acre-inches* |
| 1.19 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## 3 - 05/15/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 05/15/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 164.44 | 44.76 | 247.75 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 151.80 | 0.00 | 0.00 | 3,539.49 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 330.24 | 44.76 | 247.75 | 3,539.49 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 0.00 | 0.00 | 0.00 | 0.00 |
| Nutrient balance | 330.24 | 44.76 | 247.75 | 3,539.49 |
| Applied to removed ratio | 0.00 | 0.00 | 0.00 | 0.00 |

| Fresh water applied | |
|---|---|
| 16,965,818.10 | *gallons* |
| 624.79 | *acre-inches* |
| 34.71 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

NUTRIENT ANALYSES

## A. MANURE ANALYSES

Manure 2305225-01

Sample and source description: Manure 2305225-01

Sample date: 05/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: _____ %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 8,230.00 | 2,240.00 | 12,400.00 | | | | | | | |
| **DL** | 5.00 | 0.50 | 0.50 | | | | | | | |

## B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2301393-01

Sample and source description: Lagoon Water 2301393-01

Sample date: 01/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 204.00 | 111.00 | | | 55.00 | 402.00 | | | | | | | | 3,410.00 | 2,900 |
| **DL** | 0.60 | 6.00 | | | 0.10 | 10.00 | | | | | | | | 1.00 | 1 |

Lagoon Water 2303393-01

Sample and source description: Lagoon Water 2303393-01

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 361.00 | 361.00 | | | 93.40 | 775.00 | | | | | | | | 5,210.00 | 3,700 |
| **DL** | 0.60 | 5.00 | | | 0.10 | 10.00 | | | | | | | | 1.00 | 1 |

Lagoon Water 2307222-01

Sample and source description: Lagoon Water 2307222-01

Sample date: 07/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 192.00 | 139.00 | | | 61.00 | 307.00 | | | | | | | | 3,110.00 | 1,800 |
| **DL** | 0.60 | 0.60 | | | 0.10 | 10.00 | | | | | | | | 1.00 | 1 |

Lagoon Water 2310034-01

Sample and source description: Lagoon Water 2310034-01

Sample date: 09/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 9.46 | 9.46 | | | 14.40 | 65.20 | | | | | | | | 1,200.00 | 640 |
| **DL** | 0.60 | 0.60 | | | 0.10 | 10.00 | | | | | | | | 1.00 | 1 |

Lagoon Water 2312555-01

Sample and source description: Lagoon Water 2312555-01

Sample date: 12/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 192.00 | 140.00 | | | 55.10 | 31,900.00 | | | | | | | | 2,140.00 | 2,100 |
| **DL** | 0.60 | 1.00 | | | 0.10 | 10.00 | | | | | | | | 1.00 | 1 |

## C. FRESH WATER ANALYSES

Ag Well

## Ag Well

### 2312115-01

Sample description: 2312115-01

Sample date: 12/08/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 51.50 | | 47.50 | | | | | | | | 1,720.00 | 1,500 |
| **DL** | 1.40 | | 0.40 | | | | | | | | 1.00 | 1 |

## Canal Water

### 2310035-01

Sample description: 2310035-01

Sample date: 09/30/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 19.30 | | 18.90 | | | | | | | | 733.00 | 450 |
| **DL** | 0.40 | | 0.40 | | | | | | | | 1.00 | 1 |

## Domestic Well

### 2312114-01

Sample description: 2312114-01

Sample date: 12/08/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 7.06 | | | | | | | | | | 994.00 | |
| **DL** | 0.40 | | | | | | | | | | 1.00 | |

**D. SOIL ANALYSES**

*No soil analyses entered.*

## E. PLANT TISSUE ANALYSES

1 - 11/25/2022: Winter Forage

2305226-01

Sample and source description: 2305226-01

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 25,400.00 | 2,590.00 | 15,400.00 | | 17.00 |
| **DL** | 2.50 | 0.50 | 0.50 | | 0.10 |

## F. SUBSURFACE (TILE) DRAINAGE ANALYSES

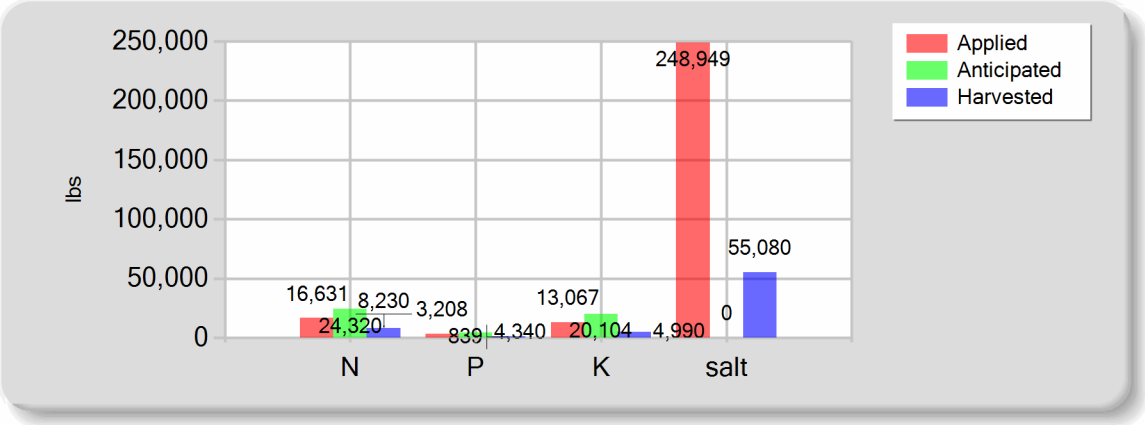
*No subsurface (tile) drainage analyses entered.*

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

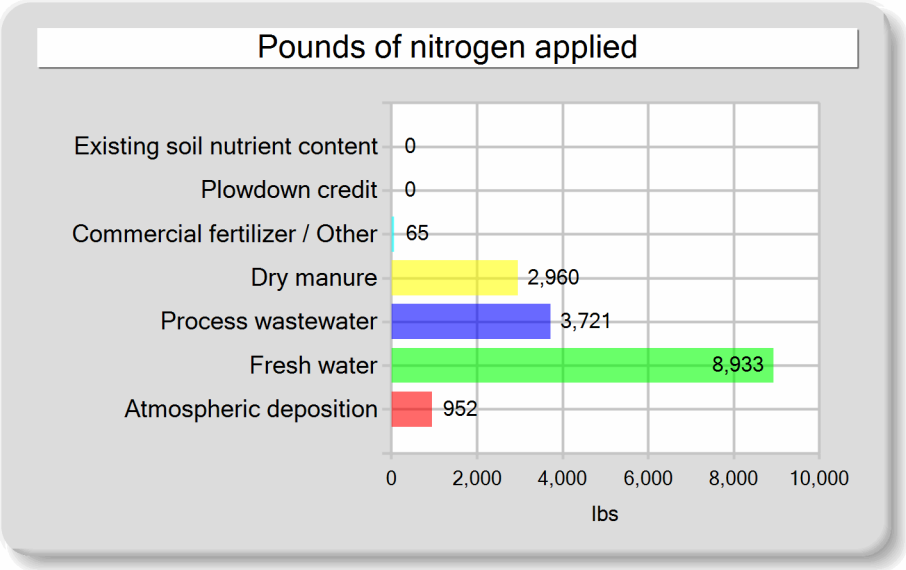
## A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

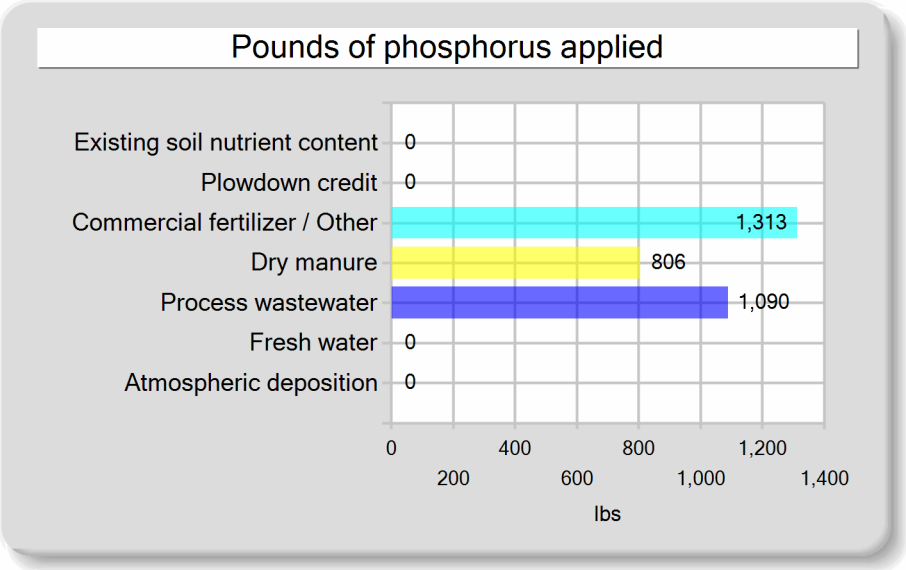
| | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 65.00 | 1,312.50 | 1,312.50 | 0.00 |
| Dry manure | 2,959.84 | 805.59 | 4,459.54 | 0.00 |
| Process wastewater | 3,721.36 | 1,089.97 | 7,295.06 | 38,506.63 |
| Fresh water | 8,933.12 | 0.00 | 0.00 | 210,442.57 |
| Atmospheric deposition | 952.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 16,631.31 | 3,208.07 | 13,067.10 | 248,949.20 |
| Anticipated crop nutrient removal | 24,320.00 | 4,340.00 | 20,104.00 | 0.00 |
| Actual crop nutrient removal | 8,229.60 | 839.16 | 4,989.60 | 55,080.00 |
| Nutrient balance | 8,401.71 | 2,368.91 | 8,077.50 | 193,869.20 |
| Applied to removed ratio | 2.02 | 3.82 | 2.62 | 4.52 |

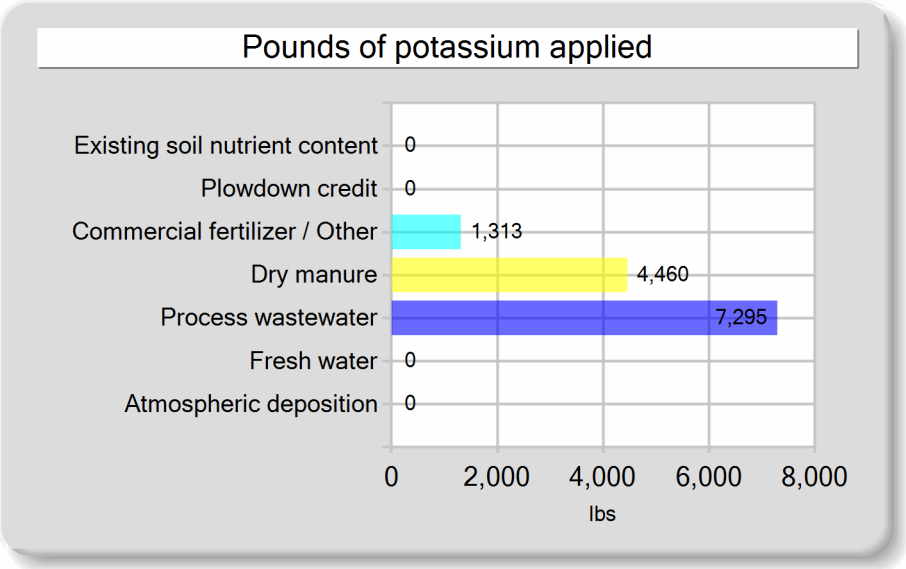
## B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

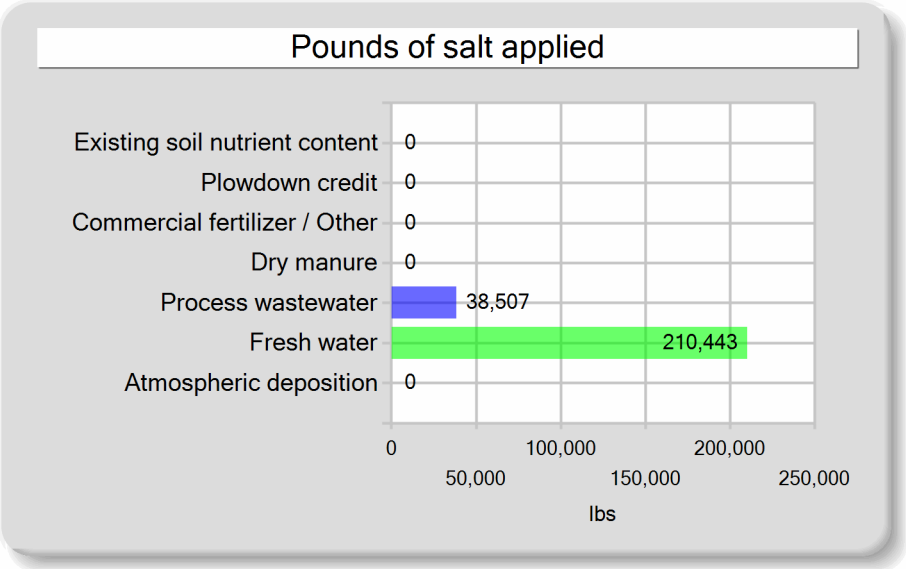


## C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE



## EXCEPTION REPORTING

### A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

*No manure or process wastewater discharges occurred during the reporting period.*

### B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

*No stormwater discharges occurred during the reporting period.*

### C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

*No land application area to surface water discharges occurred during the reporting period.*

## NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

### A. NUTRIENT MANAGEMENT PLAN STATEMENTS

| | |
|---|---|
| Was the facility's NMP updated in the reporting period? | No |
| Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |
| Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |

### B. EXPORT AGREEMENT STATEMENT

| | |
|---|---|
| Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period? | No |

ADDITIONAL NOTES

**A. NOTES**

*Field #1 & #3 Corn Silage Plant Tissue was not sampled in 2023, we will be sure to sample in 2024.

**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

CERTIFICATION

**A. OWNER AND/OR OPERATOR CERTIFICATION**

*I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.*

| | |
|---|---|
| X | X |
| SIGNATURE OF OWNER OF FACILITY | SIGNATURE OF OPERATOR OF FACILITY |
| A&N 1999 Trust-Survivors Trust, John Zylstra | John Zylstra |
| PRINT OR TYPE NAME | PRINT OR TYPE NAME |
| ✓ 6-28-24 | ✓ 6-28-24 |
| DATE | DATE |

ATTACHMENTS

## A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

| Annual Dairy Facility Assessment | N/A |
|---|---|
| Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A ) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed . | |

| Manure/Process Wastewater Tracking Manifests | N/A |
|---|---|
| Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler. | |

| Corrective Actions Documents | N/A |
|---|---|
| Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction. | |

| Groundwater Monitoring | N/A |
|---|---|
| Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13. | |

| Storm Water Monitoring | N/A |
|---|---|
| Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14. | |